

Vector Calculus (Marsden, Tromba) Ch. 1

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Complete the computations in exercises 1 to 4

1. $(-21, 23) - (?, 6) = (-25, ?)$

For questions of this sort, we break the vector into components:

$$-21 - ? = -25$$

$$? = 4$$

Now the second component:

$$23 - 6 = ?$$

$$? = 17$$

$$(-21, 23) - (4, 6) = (-25, 17)$$

2. $3(133, -0.33, 0) + (-399, 0.99, 0) = (?, ?, ?)$ We follow the rules of scalar multiplication and vector addition.

$$3(133, -0.33, 0) + (-399, 0.99, 0) = (0, 0, 0)$$

3. $(8a, -2b, 13c) = (52, 12, 11) + \frac{1}{2}(?, ?, ?)$ Once again, we solve each component separately:

$$8a = 52 + \frac{1}{2}?$$

$$16a - 104 = ?$$

y -component:

$$-2b = 12 + \frac{1}{2}?$$

$$-4b - 24 = ?$$

z -component:

$$13c = 11 + \frac{1}{2}?$$

$$26c - 22 = ?$$

Thus,

$$(8a - 2b, 13c) = (52, 12, 11) + \frac{1}{2}(16a - 104, -4b - 24, 26c - 22)$$

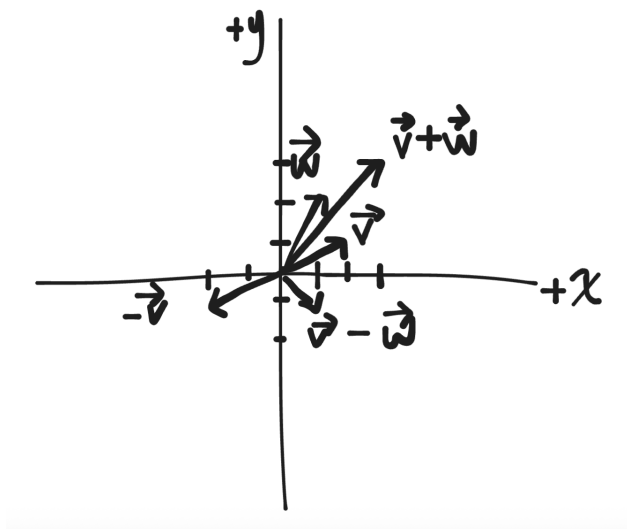
4. $(2, 3, 5) - 4\mathbf{i} + 3\mathbf{j} = (?, ?, ?)$

We know $\mathbf{i} = (1, 0, 0)$ and $\mathbf{j} = (0, 1, 0)$, so after substituting those vectors in we get

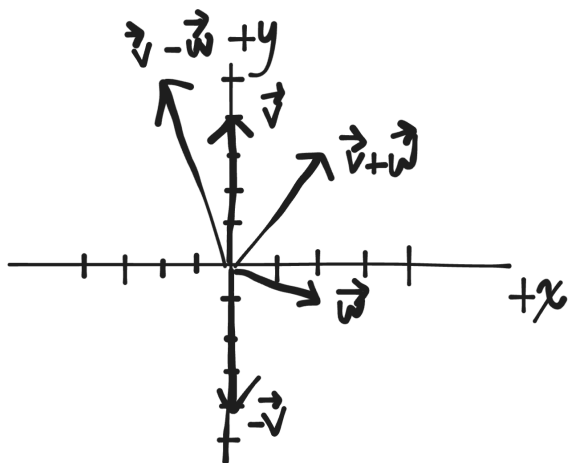
$$(2, 3, 5) - 4\mathbf{i} + 3\mathbf{j} = (-2, 6, 5)$$

In Exercises 5 to 8, sketch the given vectors $\mathbf{v}$ and $\mathbf{w}$. On your sketch, draw in $-\mathbf{v}$, $\mathbf{v} + \mathbf{w}$, $\mathbf{v} - \mathbf{w}$.

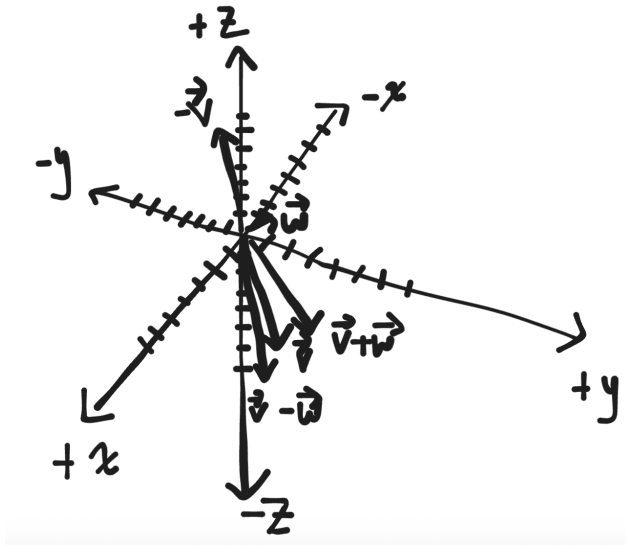
5. $\mathbf{v} = (2, 1)$ and $\mathbf{w} = (1, 2)$



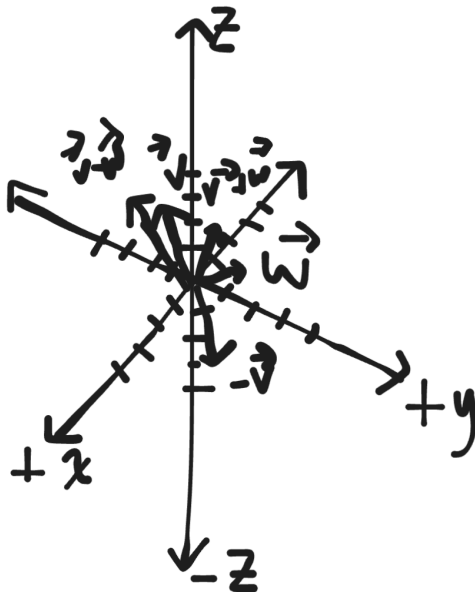
6. $\mathbf{v} = (0, 4)$ and $\mathbf{w} = (2, -1)$



7. $\mathbf{v} = (2, 3, -6)$ and $\mathbf{w} = (-1, 1, 1)$



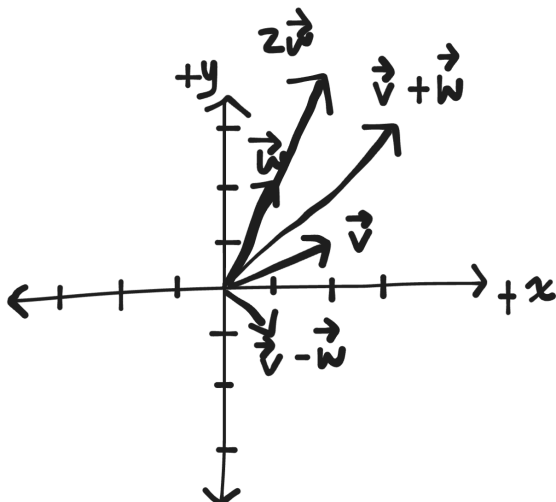
8. $\mathbf{v} = (2, 1, 3)$ and $\mathbf{w} = (-2, 0, -1)$



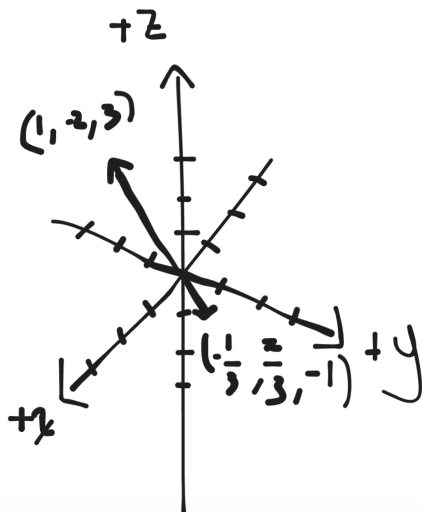
9. Let $\mathbf{v} = 2\mathbf{i} + \mathbf{j}$ and $\mathbf{w} = \mathbf{i} + 2\mathbf{j}$. Sketch $\mathbf{v}$, $\mathbf{w}$, $\mathbf{v} + \mathbf{w}$, $2\mathbf{w}$, and $\mathbf{v} - \mathbf{w}$ in the plane.

$$\mathbf{v} = 2\mathbf{i} + \mathbf{j} = 2(1, 0) + (0, 1) = (2, 1)$$

$$\mathbf{w} = \mathbf{i} + 2\mathbf{j} = (1, 0) + 2(0, 1) = (1, 2)$$



10. Sketch $(1, -2, 3)$ and $(-\frac{1}{3}, \frac{2}{3}, -1)$. Why do these vectors point in opposite directions?



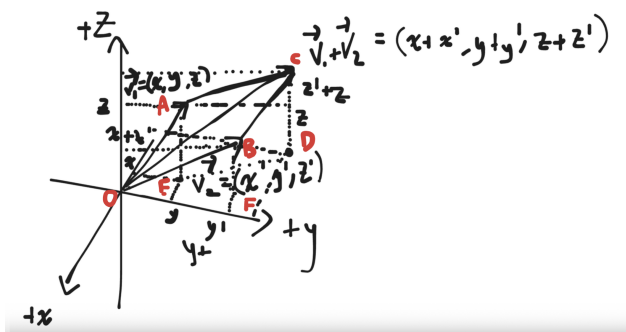
These vectors point in opposite directions because they are negative scalar multiples of each other:

$$(1, -2, 3) = -3(-\frac{1}{3}, \frac{2}{3}, -1)$$

11. What restrictions on x, y , and z so that the triple (x, y, z) will represent a point on the y axis? On the z axis? In the xz plane? In the yz plane?

For (x, y, z) to lie on the y axis, $x = 0$ and $z = 0$ must hold. For the point to be on the z axis, $x = 0$ and $y = 0$ must hold. For the xz plane, $y = 0$ must hold. For the yz plane, $x = 0$ must hold.

12. (a) Generalize the geometric construction in Figure 1.1.7 to show that if $\mathbf{v}_1 = (x, y, z)$ and $\mathbf{v}_2 = (x', y', z')$, then $\mathbf{v}_1 + \mathbf{v}_2 = (x + x', y + y', z + z')$.

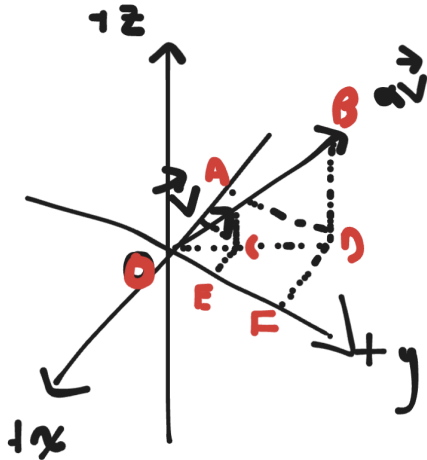


Here, we let A be (x, y, z) , B be (x', y', z') . We then draw E , F , and D , which are A , B , and C respectively projected onto the xy plane. In other words, $E = (x, y, 0)$, $F = (x', y', 0)$, and $G = (x + x', y + y', 0)$.

From the two dimensional case, we know that adding vectors E and F tip to tail results in the same vector as $(x + x', y + y')$. Thus, adding A and B tip to tail guarantees that the result will x and y coordinates equal to $x + x'$ and $y + y'$ respectively.

It remains to show that adding A and B tip to tail also provides the correct z component. We can simply project the points onto the xz plane to confirm that the z value of adding A and B tip to tail coincides with C .

- (b) Using an argument based on similar triangles, prove that $\alpha \mathbf{v} = (\alpha x, \alpha y, \alpha z)$ when $\mathbf{v} = (x, y, z)$.



Here, A is the point that corresponds to vector $\mathbf{v}$ and B the point that corresponds to vector $\alpha \mathbf{v}$. Since $\overline{AC}$ and $\overline{BD}$ are parallel, $\angle OAC = \angle OBD$ and $\angle OCA = \angle ODB$. Furthermore, since they are the same angle, $\angle AOC = \angle BOD$. Thus, $\triangle AOC \sim \triangle BOD$ by AAA similarity..

We are given that the vector B is α times longer than A , so the scale factor between the two similar triangles is α . Thus, $\alpha \overline{AC} = \overline{BD}$, so the z -coordinate of B is indeed αz .

We also get that $\alpha \overline{OC} = \overline{OD}$. Because $\overline{CE} \parallel \overline{DF}$, $\angle OCE = \angle ODF$ and $\angle OEC = \angle OFD$. Also, $\angle COE = \angle DOF$. We conclude that $\triangle OCE \sim \triangle ODF$ and because $\alpha \overline{OC} = \overline{OD}$, the scale factor is α .

We can then derive that $\alpha\overline{OE} = \overline{OF}$ and $\alpha\overline{EC} = \overline{DF}$, and since the length of $\overline{OF}$ is $\alpha\mathbf{v}$'s y -coordinate and the length of $\overline{DF}$ is $\alpha\mathbf{v}$'s x -coordinate we have shown that $\alpha\mathbf{v} = (\alpha x, \alpha y, \alpha z)$.

In Exercises 13 to 19, use set theoretic or vector notation or both to describe the points that lie in the given configurations.

13. The plane spanned by $\mathbf{v}_1 = (2, 7, 0)$ and $\mathbf{v}_2 = (0, 2, 7)$

The set of vectors of the form $s\mathbf{v}_1 + t\mathbf{v}_2$ where s and t are real numbers.

14. The plane spanned by $\mathbf{v}_1 = (3, -1, 1)$ and $\mathbf{v}_2 = (0, 3, 4)$

The set of vectors of the form $s\mathbf{v}_1 + t\mathbf{v}_2$ where s and t are real numbers.

15. The line passing through $(-1, -1, -1)$ in the direction of $\mathbf{j}$

The set of vectors of the form $\mathbf{j} + s(-1, -1, -1)$ where s is a real number.

16. The line passing through $(0, 2, 1)$ in the direction of $2\mathbf{i} - \mathbf{k}$

The set of vectors of the form $(0, 2, 1) + s(2\mathbf{i} - \mathbf{k})$ where s is a real number.

17. The line passing through $(-1, -1, -1)$ and $(1, -1, 2)$

Since $(-1, -1, -1) - (1, -1, 2) = (-2, 0, -3)$, our answer is the set of vectors of the form $(-1, -1, -1) + s(-2, 0, -3)$ where s is a real number.

18. The line passing through $(-5, 0, 4)$ and $(6, -3, 2)$

Since $(-5, 0, 4) - (6, -3, 2) = (-11, 3, 2)$, our answer is the set of vectors of the form $(-5, 0, 4) + s(-11, 3, 2)$, where s is a real number.

19. The parallelogram whose adjacent sides are the vectors $\mathbf{i} + 3\mathbf{k}$ and $-2\mathbf{j}$

The set of vectors of the form $s(\mathbf{i} + 3\mathbf{k}) + t(-2\mathbf{j})$ where $0 \leq s, t \leq 1$.

20. Show that $\mathbf{l}_1(t) = (1, 2, 3) + t(1, 0, -2)$ and $\mathbf{l}_2(t) = (2, 2, 1) + t(-2, 0, 4)$ parametrize the same line.

Let $\mathbf{v} = (1, 2, 3) + t(1, 0, -2)$. Then, we wonder if there exists t_2 such that $\mathbf{l}_2(t_2) = \mathbf{v}$. We have 3 equations (one for each component) we must satisfy:

$$2 - 2t_2 = 1 + t$$

$$2 = 2$$

$$1 + 4t_2 = 3 - 2t$$

The second equation is always true, so we only worry about the other 2. Isolating t in the first equation gives us

$$t = 1 - 2t_2$$

We substitute this into the second equation

$$1 + 4t_2 = 3 - 2 + 4t_2$$

$$1 = 1$$

We can see that $t = 1 - 2t_2$ always satisfies the equation, which is equivalent to $t_2 = \frac{1-t}{2}$.

Thus, if $\mathbf{v} = (1, 2, 3) + t(1, 0, -2)$, then $\mathbf{l}_2(\frac{1-t}{2}) = \mathbf{v}$. Likewise, if $\mathbf{v} = (2, 2, 1) + t_2(-2, 0, 4)$, then $\mathbf{l}_1(1 - 2t_2) = \mathbf{v}$. We can conclude the members of both sets are the same, meaning the two parametrized lines are the same.

21. Do the points $(2, 3, -4)$, $(2, 1, -1)$, and $(2, 7, -10)$ lie on the same line?

First, we arrive at a possible equation for the line between points $(2, 3, -4)$ and $(2, 1, -1)$:

$$\mathbf{y}(t) = (2, 3, -4) + t((2, 1, -1) - (2, 3, -4))$$

$$\mathbf{y}(t) = (2, 3, -4) + t(0, -2, 3)$$

Now, we verify whether $(2, 7, -10)$ also lies on this line. We can try to solve for t that satisfies the equations

$$2 = 2$$

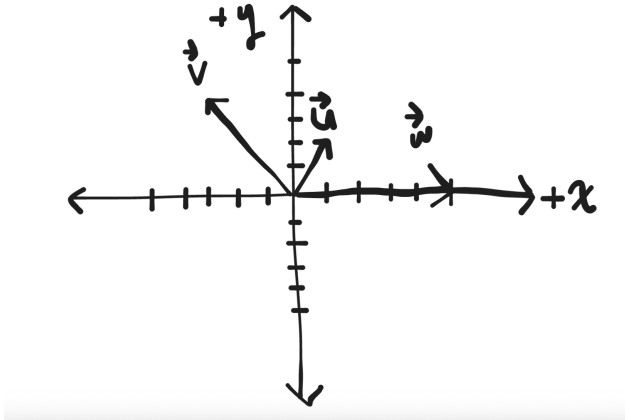
$$3 - 2t = 7$$

$$-4 + 3t = -10$$

The second equation gives $t = -2$, which satisfies the other two equations. We conclude that all three points lie on the same line.

22. Let $\mathbf{u} = (1, 2)$, $\mathbf{v} = (-3, 4)$, and $\mathbf{w} = (5, 0)$.

- (a) Draw these vectors in R^2 .



- (b) Find scalars λ_1 and λ_2 such that $\mathbf{w} = \lambda_1\mathbf{u} + \lambda_2\mathbf{v}$.

Each component gives us an equation:

$$\lambda_1 - 3\lambda_2 = 5$$

$$2\lambda_1 + 4\lambda_2 = 0$$

The first equation gives $\lambda_1 = 5 + 3\lambda_2$. We substitute this into the second:

$$10 + 6\lambda_2 + 4\lambda_2 = 0$$

$$10 = -10\lambda_2$$

$$\lambda_2 = -1$$

Then, we can substitute back into the first equation to get $\lambda_1 = 2$, so our solution is $\lambda_1 = 2$ and $\lambda_2 = -1$.

23. Suppose A , B , and C are vertices of a triangle. Find $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA}$.

Since adding vectors simply means adding them tip to tail, we realize that performing addition in the given order takes us to the original point we start at, so the answer is $\mathbf{0}$.

24. Find the points of intersection of the line $x = 3 + 2t$, $y = 7 + 8t$, $z = -2 + t$, that is, $\mathbf{l}(t) = (3 + 2t, 7 + 8t, -2 + t)$ with the coordinate planes.

For a point to lie on the xy plane, the z coordinate must be 0. Thus, $0 = -2 + t$ gives $t = 2$, and $\mathbf{l}(2) = (7, 23, 0)$ is the intersection of this line with the xy plane.

For the xz plane, we require $y = 0$. Thus, $0 = 7 + 8t$, so $t = -\frac{7}{8}$. $\mathbf{l}(-\frac{7}{8}) = (\frac{5}{4}, 0, -\frac{23}{8})$.

Lastly, for the yz plane, $x = 0$ must hold. This furnishes $t = -\frac{3}{2}$, so $\mathbf{l}(-\frac{3}{2}) = (0, -5, -\frac{7}{2})$.

25. Show that there are no points (x, y, z) satisfying $2x - 3y + z - 2 = 0$ and lying on the line $\mathbf{v} = (2, -2, -1) + t(1, 1, 1)$.

For a point (x, y, z) to be on the line $\mathbf{v}$, there must exist t satisfying the following equations:

$$x = 2 + t$$

$$y = -2 + t$$

$$z = -1 + t$$

From this we can deduce that $y = x - 4$ and $z = x - 3$. We can then write $2x - 3y + z - 2 = 0$ in terms of x :

$$2x - 3(x - 4) + x - 3 = 0$$

$$2x - 3x + 12 + x - 3 = 0$$

$$9 = 0$$

Clearly, this is impossible. Thus, we have shown that no point can satisfy both conditions as if it lies on the line $\mathbf{v}$, then it can't satisfy $2x - 3y + z - 2 = 0$.

26. Show that every point on the line $\mathbf{v} = (1, -1, 2) + t(2, 3, 1)$ satisfies the equation $5x - 3y - z - 6 = 0$.

Let (x, y, z) be a point on line $\mathbf{v}$. Then, there exists t such that:

$$x = 1 + 2t$$

$$y = -1 + 3t$$

$$z = 2 + t$$

We substitute t into the other condition:

$$5(1 + 2t) - 3(-1 + 3t) - (2 + t) - 6 = 0$$

$$5 + 10t + 3 - 9t - 2 - t - 6 = 0$$

$$6 - 6 = 0$$

$$0 = 0$$

Thus, since (x, y, z) is a point on line $\mathbf{v}$, x, y, z also satisfy the given equation.

27. Determine whether the lines $x = 3t + 2$, $y = t - 1$, $z = 6t + 1$ and $x = 3s - 1$, $y = s - 2$, $z = s$ intersect.

We want to see if there exists s and t such that all the components are equal. In other words:

$$3t + 2 = 3s - 1$$

$$t - 1 = s - 2$$

$$6t + 1 = s$$

The third equation gives us $s = 6t + 1$. Thus, substituting into the second equation gives

$$t - 1 = 6t - 1$$

$$5t = 0$$

$$t = 0$$

$s = 1$ and $t = 0$ satisfy all 3 equations, so the lines do indeed intersect.

28. Do the lines $(x, y, z) = (t + 4, 4t + 5, t - 2)$ and $(x, y, z) = (2s + 3, s + 1, 2s - 3)$ intersect? We use the same strategy as problem 27:

$$t + 4 = 2s + 3$$

$$4t + 5 = s + 1$$

$$t - 2 = 2s - 3$$

This first equation gives $t = 2s - 1$, while the third gives $t = 2s - 5$. Clearly, both of these statements cannot be true, so there are no s and t such that both lines yield the same point, meaning they don't intersect.

In Exercises 29 to 31, use vector methods to describe the given configurations.

29. The parallelepiped with edges the vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ emanating from the origin.

All the points that lie in this parallelepiped can be described by $r\mathbf{a} + s\mathbf{b} + t\mathbf{c}$ where $0 \leq r, s, t \leq 1$.

30. The points within the parallelogram with one corner at (x_0, y_0, z_0) whose sides extending from that corner are equal in magnitude and direction to vectors $\mathbf{a}$ and $\mathbf{b}$.

All points in this parallelogram can be described by $(x_0, y_0, z_0) + s\mathbf{a} + t\mathbf{b}$, where $0 \leq s, t \leq 1$.

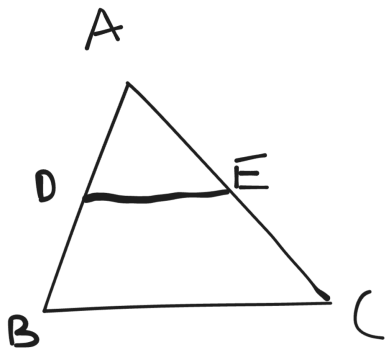
31. The plane determined by the three points (x_0, y_0, z_0) , (x_1, y_1, z_1) , and (x_2, y_2, z_2)

We want to find the vectors that span this plane. By subtracting the vectors corresponding to these points, we will arrive at possible vectors that span the plane. For example, take $\mathbf{v} = (x_1 - x_0, y_1 - y_0, z_1 - z_0)$ and $\mathbf{u} = (x_2 - x_0, y_2 - y_0, z_2 - z_0)$.

Then, all vectors of the form $s\mathbf{v} + t\mathbf{u}$ make up the desired plane, where s and t are real numbers.

Prove the statements in Exercises 32 to 34.

32. The line segment joining the midpoints of two sides of a triangle is parallel to and has half the length of the third side.



In this image, we can see $\overline{DE}$ is the line segment joining the midpoints of two sides of a triangle and $\overline{BC}$ is the third side.

If $\overrightarrow{DE}$ is the vector from D to E and $\overrightarrow{BC}$ is the vector from B to C and we can show $2\overrightarrow{DE} = \overrightarrow{BC}$, then we will have proved that both vectors are the same direction (meaning they are parallel) and $\overline{DE}$ is half the length of $\overline{BC}$.

Let $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ be the vectors corresponding to points A , B and C of the triangle, respectively. Since D is the midpoint of $\overline{AB}$, the vector corresponding to point D is $(\frac{a_1+b_1}{2}, \frac{a_2+b_2}{2}, \frac{a_3+b_3}{2})$. Similarly, the vector corresponding to E is $(\frac{a_1+c_1}{2}, \frac{a_2+c_2}{2}, \frac{a_3+c_3}{2})$.

We can subtract these vectors to get $\overrightarrow{DE} = (\frac{c_1-b_1}{2}, \frac{c_2-b_2}{2}, \frac{c_3-b_3}{2})$. The vector $\overrightarrow{BC} = (c_1 - b_1, c_2 - b_2, c_3 - b_3)$. We can see that $2\overrightarrow{DE} = \overrightarrow{BC}$, thus completing the proof.

33. If PQR is a triangle in space and $b > 0$ is a number, then there is a triangle with sides parallel to those of PQR and side lengths b times those of PQR .

Consider the vectors $\mathbf{p}$, $\mathbf{q}$, $\mathbf{r}$ corresponding to the points PQR , respectively. Let $\mathbf{u}$ be the vector $\mathbf{p} + b(\mathbf{q} - \mathbf{p})$ and $\mathbf{v}$ be the vector $\mathbf{p} + b(\mathbf{r} - \mathbf{p})$. We associate point Q' with $\mathbf{u}$ and R' with $\mathbf{v}$.

The triangle $PQ'R'$ is exactly the triangle we desire: Since $\overrightarrow{PQ} = \mathbf{q} - \mathbf{p}$ and $\overrightarrow{PQ'} = \mathbf{u} - \mathbf{p} = b(\mathbf{q} - \mathbf{p})$, $\overrightarrow{PQ'}$'s magnitude is b times that of $\overrightarrow{PQ}$ and is also parallel, since $\overrightarrow{PQ'} = b\overrightarrow{PQ}$. We can use similar logic to conclude that $\overrightarrow{PR'} = b\overrightarrow{PR}$.

Furthermore, both $\triangle PQR$ and $\triangle PQ'R'$ share the same angle $\angle QPR$, and since we have just proved that the surrounding sides are proportional, we can use *SAS* similarity to deduce that the final side $\overline{R'Q'}$ is also b times longer than $\overline{RQ}$.

And because $\angle PQR = \angle PQ'R'$ (recall the similar triangles) PQ' is a transversal intersecting parallel lines QR and $Q'R'$, proving that the final edge of the triangle is still parallel.

34. The medians of a triangle intersect at a point, and this point divides each median in a ratio of $2 : 1$.

Consider an arbitrary triangle $\triangle ABC$. Let D be the midpoint of $\overline{AB}$, E be the midpoint of $\overline{AC}$, and F be the midpoint of $\overline{BC}$.

The medians of the triangle are the line segments $\overline{DC}$, $\overline{EB}$, and $\overline{FA}$. First, we show that these three line segments intersect.

The points on the line segment $\overline{DC}$ can be described in vector form by $D + t(C - D)$, where $0 \leq t \leq 1$. Let's take $t = \frac{1}{3}$.

$$D + \frac{1}{3}(C - D)$$

Recall $D = \frac{A+B}{2}$

$$\begin{aligned} \frac{A+B}{2} + \frac{1}{3}\left(C - \frac{A+B}{2}\right) \\ \frac{A+B}{2} - \frac{A+B}{6} + \frac{C}{3} \\ \frac{A+B+C}{3} \end{aligned}$$

We can use similar logic to show that all medians intersect at the point corresponding to the vector $\frac{A+B+C}{3}$. Let this vector be M .

We can see that $M - D = \frac{A+B+C}{3} - \frac{A+B}{2} = \frac{2C-A-B}{6}$. On the other hand, $C - M = C - \frac{A+B+C}{3} = \frac{2C-A-B}{3}$. We see that the ratio between the length of the segment from the side of the triangle to the median intersection to the length of the median intersection to the opposite vertex is $1 : 2$ as desired.

Problems 35 and 36 require some knowledge of chemical notation.

35. Write the chemical equation $\text{CO} + \text{H}_2\text{O} = \text{H}_2 + \text{CO}_2$ as an equation in ordered triples (x_1, x_2, x_3) , where x_1, x_2, x_3 are the number of carbon, hydrogen, and oxygen atoms, respectively in each molecule.

$$(1, 0, 1) + (0, 2, 1) = (0, 2, 0) + (1, 0, 2)$$

36. (a) Write the chemical equation $p\text{C}_3\text{H}_4\text{O}_3 + q\text{O}_2 = r\text{CO}_2 + s\text{H}_2\text{O}$ as an equation in ordered triples with unknown coefficients p, q, r , and s .

$$p(3, 4, 3) + q(0, 0, 2) = r(1, 0, 2) + s(0, 2, 1)$$

- (b) find the smallest positive integer solution for p, q, r , and s .

We have 3 equations that p, q, r, s must satisfy:

$$3p = r$$

$$4p = 2s$$

$$3p + 2q = 2r + s$$

The first equation gives $r = 3p$, and the second gives $s = \frac{p}{2}$. Substituting this into the 3rd equation gives

$$3p + 2q = 6p + \frac{p}{2}$$

$$2q = \frac{7}{2}p$$

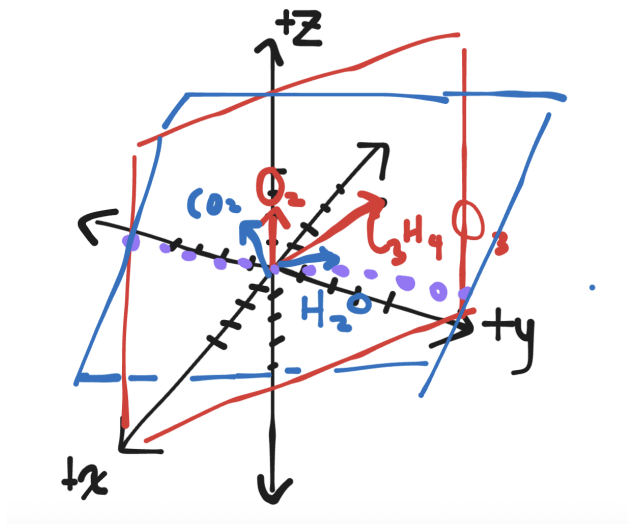
$$4q = 7p$$

We can see that the smaller that q is, the smaller p , and thus r and s , will be. Thus, minimizing q gives us the minimum positive integer solution in general. We require that $4q$ be a multiple of 7, so that p can be an integer. Thus, $q = 7$ is the smallest possible value of q . We then check if the rest of the unknowns are still positive integers.

This gives, $q = 7$, $p = 4$, $r = 12$, and $s = 2$. Since all values are positive integers, we have the minimum integer solution.

- (c) Illustrate the solution by a vector diagram in space.

The LHS of the chemical reaction equation gives a plane, as does the right. We are essentially finding the intersection of these planes, in particular the point of intersection with positive integer coordinates.



37. Find a line that lies entirely in the set defined by the equation $x^2 + y^2 - z^2 = 1$.

We can note that $x = 1$ and $y = z$, then the equation is always true. Thus, we just need to come up with a line that satisfies those properties:

$$\mathbf{v}(t) = (1, 0, 0) + t(0, 1, 1)$$